

Question Paper

Exam Date & Time: 31-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Fundamentals of Machine Learning [CSE_3127]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) With the help of a neat diagram, represent both hard-margin and soft-margin Support Vector Machines (SVMs). (5)
- A) i) Analyse how the optimization formulation changes when moving from a hard-margin SVM to a soft-margin SVM.
ii) Explain the role of slack variables (ξ_i) and the regularization parameter (C) in controlling margin violations.
iii) Support your answer with a suitable illustrative example showing how varying C affects the decision boundary and misclassifications.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Explain the process of Boosting using a diagram and stepwise flow. Analyse how successive weak learners reduce bias and improve overall model accuracy. (3)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- C) **For the matrix** (2)
- $$A = \begin{bmatrix} 3 & 0 \\ 0 & 0 \end{bmatrix},$$
- apply SVD to determine the left singular vector (u_1) associated with the non-zero singular value.**

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2) Analyse how the structure of a decision tree influences the form and number of IF-THEN rules that can be extracted from it. Explain how the placement of attributes near the root impacts interpretability and generalization of the resulting rule set. (4)
- A)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) Illustrate the Random Forest algorithm with a diagram and explain the key steps involved in training and combining multiple trees. Analyse how Random Forest overcomes overfitting compared to a single decision tree. (3)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- C) Analyse how the bias-variance trade-off affects a model's generalization ability. Explain how high bias and high variance manifest as underfitting and overfitting respectively, using suitable examples. (3)

3) You are given the following training data: (4)

A)

Taste	Temperature	Beverage
Sweet	Cold	Juice
Bitter	Hot	Coffee
Bitter	Cold	ColdCoffee
Sweet	Hot	Tea
Sweet	Cold	Juice

A new beverage sample has Taste = Sweet and Temperature = Hot. Apply the Naïve Bayes classifier to find the most probable BeverageType.

B) Analyse how the Winnow algorithm's multiplicative weight-update strategy makes it more effective for high-dimensional sparse datasets compared to the Perceptron. (3)

C) A logistic regression model estimates purchase probability: (3)

$$P(\text{Buy}) = \frac{1}{1 + e^{-(2 + 0.7x_1 + 0.02x_2)}}$$

where

x_1 = Advertisement Exposure (hours/week)

x_2 = Product Discount (%)

For a customer exposed 5 hours/week to ads and given a 10% discount,

1. Compute $P(\text{Buy})$.
2. Predict if they will buy when threshold = 0.65.
3. Re-evaluate if the discount increases to 15% while ad exposure remains the same.

4) (5)

A)

You are classifying animals based on BodyLength (cm) and TailLength (cm).

ID	BodyLength (x)	TailLength (y)	Species
A	20	5	Rabbit
B	28	11	Fox
C	78	2	Wolf
D	25	6	Rabbit
E	60	18	Wolf
F	36	9	Fox

Two test animals are observed:

1. $T_1 = (30, 8)$
2. $T_2 = (55, 16)$

Apply the k-NN classification algorithm with Euclidean distance and $k = 3$ to predict the species of each test animal. Show all distance calculations and the final class for T_1 and T_2 .

Course Outcome: "CO5" / Cognitive Level: "Application"

- B) Analyse how the Black Swan Paradox challenges the statistical assumption of stationarity in predictive models. Determine the reason why models trained on stable historical data fail when confronted with rare, high-impact events? (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- C) **The kernel trick avoids explicit computation of $\phi(x)$ yet achieves the same effect as transforming data into higher dimensions. Analyze the theoretical basis behind this equivalence** (2)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 5) Analyse how the mutual reinforcement between hub and authority pages in the HITS algorithm leads to score convergence over successive iterations, and examine the key computational steps involved in updating hub and authority values. (4)
- A)

Course Outcome: "CO6" / Cognitive Level: "Analysis"

- B) Analyse the key challenges faced by modern Recommendation Systems. Based on these challenges, explain how the main types of recommendation systems, Content-Based, Collaborative Filtering, address these issues differently. (3)

Course Outcome: "CO6" / Cognitive Level: "Analysis"

- C) Analyse how Bibliographic Coupling captures the relationship between documents through shared references. (3)

Course Outcome: "CO6" / Cognitive Level: "Analysis"

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Event Name	Subject
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#	Course Outcome	Description	Marks
1	1	CO 1	2
2	2	CO 2	4
3	3	CO 3	12
4	4	CO 4	11
5	5	CO 5	11
6	6	CO 6	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	CSE_3127:Fundamentals of Machine Learning

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	4	14(28%)
4	Analysis	11	36(72%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)